AMAN RAWAT

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Education

Graphic Era Hill University, Dehradun **2023 - 2027**
B.Tech in Computer Science & Engineering *7.00/10 CGPA*

Projects

Ride Sharing System (Node.js, Express.js, MongoDB, HTML, CSS, JavaScript **)** **Source Code**

- Developed a **full-stack peer-to-peer Ride Sharing web application** for GEHU students **managing 50+ ride** entries with MongoDB-backed data storage and 14 RESTful APIs for authentication, ride posting and booking.
- Implemented secure login with bcrypt password hashing, institutional email restriction (@gehu.ac.in) and role-based CRUD operations for Riders (post/edit/delete rides) and Seaters (search/book rides).
- Designed a **responsive Single Page Application with dark mode**, toast notifications and mobile-friendly UI with Git version control on GitHub.

DSA Visualiser (Java, HTML, CSS, JavaScript, Data Structures, Algorithms**)** **Source Code**

- Developed an **interactive DSA visualisation tool** to demonstrate algorithms through real-time animations for better understanding.
- Implemented visual representations of **sorting and searching algorithms** such as Bubble Sort, Merge Sort, and Binary Search.
- Designed step-by-step execution flow to help users grasp **time complexity and algorithm behavior** intuitively.
- Built a responsive interface enabling users to **control speed, input data, and visualize algorithm operations dynamically**.

Static Code Analyser (Python, Compiler Design, Lexical Analysis, Parsing **)** **Source Code**

- Developed a **static code analysis tool** to examine source code without execution, identifying syntax errors, inefficiencies, and potential vulnerabilities.
- Implemented core **compiler design concepts** such as lexical analysis, tokenization, and parsing to analyze code structure.
- Built modules to detect **code quality issues**, improper syntax patterns, and maintainability concerns.
- Designed a modular system to process input code and generate **detailed analysis reports** for better debugging and optimization.

AI-Powered Real-Time Chat App (PHP, MySQL, HTML, CSS, JavaScript, OpenAI API) **Source Code**

- Developed a full-stack real-time chat application supporting multi-user private messaging with JavaScript polling, online/offline status and unread message count using PHP backend and MySQL with normalized schema of 3 tables and REST API endpoints.
- Implemented secure authentication system with bcrypt password hashing, **session management and role-based access with CRUD operations** for users, messages and chat history.
- Integrated **OpenAI API** for AI-powered smart reply suggestions and built responsive **WhatsApp-style UI** with emoji support, last seen feature and live message updates without page refresh.

Technical Skills

Languages: Java, JavaScript , PHP , Python
Frontend: HTML, CSS, PHP, React
Backend: Node.js, Express.js, PHP
Databases: MySQL , MongoDB
Developer Tools: VS Code, Git, GitHub, XAMPP

Certifications

- CCNA Foundations – Networking Basics and Cisco IOS **(Coursera)** **View Certificate**
- Network Architecture Fundamentals **(Coursera)** **View Certificate**
- Data Structures And Algorithm Using JAVA **(Coursera)** **View Certificate**

Algorithmic Competitions/Achievements

- Solved **250+ DSA problems** from **Rapid Growth Problemset**, building problem solving skills.
- **Awarded NSS "B" Certificate** with **240+ hours** of community service.
- Ranked 38th out of 1500+ participants in **CodeChef** DSA Competition, placing in **top 3%** — demonstrating strong command of data structures and algorithms under time constraints.